

Rapid Revision Summary

Last-minute review · condensed from 19 papers and mark schemes, Jan 2020 – Jun 2026

THE PAPER

1 h 30 min · 75 marks · 9–10 questions · $\approx$ 1.2 min per mark.
On almost every paper: functions, trig, differentiation (19/19); models, log graphs, integration, modulus (18); iteration (17).

TOP 12 MARK-LOSERS

- “Calculator technology not acceptable”: no working = **no marks**, even if correct.
- “Hence” = use the previous part; starting afresh scores 0.
- Radians when asked (α in degrees = A0). Calculator mode for iterations and sign changes.
- R **exact** ($\sqrt{17}$, not 4.12). Reject $R = -25$.
- Sign change: two values + “change of sign” + “continuous” + “so root”.
- Keep trailing zeros: 0.8110, not 0.811.
- $f^{-1}(x)$ in terms of **x**, not “y = ...”. Range in $f(x)$ or y, **never x**.
- Do not reject valid roots ($x = 0$). No extra unrejected answers.
- $\ln|x - 5|$ with modulus; + c on indefinite; no $\int$ or + c on a definite answer.
- Differentiate **before** substituting t.
- Quotient rule $(vu' - uv')/v^2$, right way round.
- Exact when “exact”; decimal when asked (exact only = A0).

ALGEBRAIC FRACTIONS

- Factorise, cancel, common denominator $\rightarrow$ “show $f(x) = 7x/(x + 3)$ ”.
- Improper $\rightarrow$ divide first: $(2x^2 + 5x - 8)/(x + 2) = 2x + 1 - 10/(x + 2)$.
- Quadratic divisor: $f \equiv Ax^2 + Bx + C + (Dx + E)/(\text{divisor})$; compare coefficients.

FUNCTIONS

RULES

$fg(x)$: g first, then f.

f^{-1} exists only if one-to-one. Domain $f^{-1} = \text{range } f$; range $f^{-1} = \text{domain } f$.

Find f^{-1} : $y = f(x) \rightarrow$ make x the subject $\rightarrow$ swap.

$y = f^{-1}(x)$ is the reflection of $y = f(x)$ in $y = x$. Increasing f: $f = f^{-1} \Leftrightarrow f(x) = x$.

Increasing: $f'(x) > 0$ for all x in domain $\rightarrow$ state “f is increasing”.

- Jun 25: $f = 2x/(3x + 1)$, $x \geq 0 \rightarrow 0 \leq f(x) < 2/3$; $f^{-1}(x) = x/(2 - 3x)$; $f = f^{-1} \rightarrow x = 0, 1/3$.
- Jan 22: $f = (5x - 3)/(x - 4)$, $x > 4$; $f' = -17/(x - 4)^2 < 0$; $ff = (22x - 3)/(x + 13)$, range $5 < ff < 22$.

- $gf(a) = k$: $f(a) = g^{-1}(k)$. Apply inverses in reverse order.

MODULUS AND TRANSFORMATIONS

Graph	(p, q) $\rightarrow$
$f(x + b)$	$(p - b, q)$
$af(x)$	(p, aq)
$f(ax)$	$(p/a, q)$
$af(x + b)$	$(p - b, aq)$
$ f(x) $	reflect parts below x-axis
$f(x)$	reflect $x \geq 0$ part in y-axis

SOLVING |...|

Sketch $\rightarrow$ solve both cases $\rightarrow$ **check each against the sketch** $\rightarrow$ reject invalid.

Jun 24: $2|x - 5| + 10 > 6x$: $x \geq 5$ gives $x < 0$ (reject); $x < 5$ gives $x < 2.5$. Answer $x < 2.5$.

- Roots of $f(x) = k$: read heights from vertex and endpoints.
- Line through origin meets V once/twice: compare gradient with arms; vertex is the critical case.

EXPONENTIALS AND LOGS

LAWS

$e^{\ln a} = a \cdot \ln(ab) = \ln a + \ln b \cdot \ln(a/b) = \ln a - \ln b \cdot k \ln a = \ln a^k$

$Ae^{kt} = B \Rightarrow t = (1/k) \ln(B/A)$. Use $\ln$, not $\log_{10}$, to find k.

Model part	Do
start	$t = 0, e^0 = 1$
long term	$e^{-kt} \rightarrow 0$ or divide by e^{kt}
solve for t	clear fraction $\rightarrow$ isolate $e^{kt} \rightarrow \ln$
rate	differentiate, then substitute
“never reaches”	limit below the value

- Jan 20: $N = 900e^{0.12t}/(2e^{0.12t} + 1)$: $N(0) = 300$, limit 450, $N = 420 \Rightarrow t = \ln 7/0.12 = 16.22$.
- Jan 21: $\theta = A - 180e^{-kt}$: $A = 198$, $k = (1/5) \ln(5/3)$; $\theta(9) = 126^\circ\text{C}$; rate 7.33°C/min .
- Jun 21: 7.675 years = 7 years 8 months.

LOG GRAPHS

$\log y$ vs x : $y = ab^x$; intercept $\log a$, gradient $\log b$.

$\log y$ vs $\log x$: $y = ax^n$; intercept $\log a$, gradient n.

Jun 25: $\log y = -(4/7) \log x - 2 \Rightarrow y = 0.01x^{-4/7}$.

- a (or p): initial value / value at $x = 1$, **in context**.
- b: “increases by 19% each year”; 0.986: “falls by 1.4% each year”. Not “rate” or “gradient”.

TRIG I: SEC, COSEC, COT

IDENTITIES

$$\sin^2 + \cos^2 = 1 \cdot 1 + \tan^2 = \sec^2 \cdot 1 + \cot^2 = \operatorname{cosec}^2$$

$$|\sec|, |\operatorname{cosec}| \geq 1: \text{reject } \operatorname{cosec} \theta = -2/3.$$

$$\arcsin [-\pi/2, \pi/2] \cdot \arccos [0, \pi] \cdot \arctan (-\pi/2, \pi/2)$$

- $2 \sec^2 x - 3 \tan x = 2 \Rightarrow \tan x(2 \tan x - 3) = 0 \Rightarrow x = 0.983, \pi.$
- $8 \cos \theta = 3 \operatorname{cosec} \theta \Rightarrow \sin 2\theta = 3/4 \Rightarrow \theta = 24.3^\circ.$
- Factorise, never divide by $\sin/\cos$. Multiple angles: widen the range first, then halve.
- All solutions in range, no extras.

TRIG II: COMPOUND ANGLES

MUST KNOW

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

$$\tan 2A = 2 \tan A / (1 - \tan^2 A)$$

$$\sin 3x = 3 \sin x - 4 \sin^3 x \text{ (from } \sin(2x + x))$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \cdot \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

- Proof: start one side, finish at the other. $\cos 2x/\sin x + \sin 2x/\cos x = \cos(2x - x)/(\sin x \cos x) = \operatorname{cosec} x.$
- Oct 24: $2 \sin 3\theta = 5 \sin 2\theta \Rightarrow s = 0$ or $4c^2 - 5c - 1 = 0 \Rightarrow \theta = 100.1^\circ, 180^\circ, 259.9^\circ, 360^\circ.$

R FORM

$$a \cos x + b \sin x = R \cos(x - \alpha): R = \sqrt{a^2 + b^2}, \tan \alpha = b/a.$$

$$\text{Jun 25: } 7 \cos x - 24 \sin x = 25 \cos(x + 1.287).$$

$$\text{Max/min: choose } \cos = \pm 1 \text{ to make the whole expression extreme. } g = 5/(90 - 3f(2x)): \text{min } 1/33 \text{ at } 2x + 1.287 = \pi \Rightarrow x = 0.927.$$

DIFFERENTIATION

y	dy/dx	y	dy/dx
e^{kx}	ke^{kx}	$\sin kx$	$k \cos kx$
$\ln f$	f'/f	$\cos kx$	$-k \sin kx$
a^x	$a^x \ln a$	$\tan kx$	$k \sec^2 kx$
f^n	$nf f^{n-1}$	$\sec x$	$\sec x \tan x$
uv	$u'v + uv'$	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
u/v	$(vu' - uv')/v^2$	$\cot x$	$-\operatorname{cosec}^2 x$

- Stationary: $dy/dx = 0. (2x + 5)e^{3x} \rightarrow e^{3x}(6x + 17) = 0 \rightarrow x = -17/6.$
- $x = f(y): dy/dx = 1 \div dx/dy$, then convert with a Pythagorean identity. $x = 6 \sin^2 y \Rightarrow dy/dx = 1/(4\sqrt{6x - x^2}).$
- $x = ye^{2y} \Rightarrow dy/dx = y/(x(1 + 2y)).$ Vertical tangent: $dx/dy = 0.$
- Normal gradient $-1/m$; give form asked (integers, “= 0”).
- Two turning points: $dy/dx = 0$ quadratic with $b^2 - 4ac > 0.$

INTEGRATION

f(x)	$\int f dx$
e^{ax+b}	$(1/a)e^{ax+b}$
$1/(ax + b)$	$(1/a) \ln ax + b $
$(ax + b)^n$	$(ax + b)^{n+1}/(a(n+1))$
f'/f	$\ln f $
$f f^n$	$f^{n+1}/(n+1)$
$\sin(ax + b)$	$-(1/a) \cos(ax + b)$
$\cos(ax + b)$	$(1/a) \sin(ax + b)$
$\sec^2 ax$	$(1/a) \tan ax$
$\sin^2 x / \cos^2 x$	$\frac{1}{2}x \mp \frac{1}{4} \sin 2x$

- Split: $\int (x^2 - 5)/(2x^3) dx = \frac{1}{2} \ln x + 5/(4x^2) + c.$
- $\int 9x/(3x^2 + k) dx = (3/2) \ln(3x^2 + k) + c.$
- $\int \sin^3 x dx = -\frac{3}{4} \cos x + (1/12) \cos 3x + c.$
- $(1 + 2 \cos 2x)^2 = 3 + 4 \cos 2x + 2 \cos 4x$; area $0 \rightarrow 2\pi/3 = 2\pi - 3\sqrt{3}/4.$
- Exact: one log term ($p + \ln q$). Areas: curve $\pm$ triangle/trapezium.

NUMERICAL METHODS

ROOT IN [A, B] (2)

$f(a) < 0, f(b) > 0$ (values shown); change of sign and f continuous $\Rightarrow$ root in $[a, b]$. Not “the root is continuous”.

ROOT TO 3 D.P. (2)

Use bounds ± 0.0005 and **name the function**: $f(2.1535) > 0, f(2.1545) < 0 \Rightarrow \alpha = 2.154$. Iterating instead scores 0.

- $x_{n+1} = 2 \arctan(4/x_n), x_1 = 2: x_2 = 2.214, x_6 = 2.155$ (radians).
- Show every line rearranging to $x = g(x)$ (it is a “show that”).
- Full calculator value carried forward; quote to d.p. asked.

FINAL 60 SECONDS

- Mode: radians? Accuracy: 3 s.f., d.p., or exact?
- Every range in $f(x)$; every inverse in x ; every root checked against its case.
- Model: $t = 0, t \rightarrow \infty$, differentiate then substitute, interpret with % and context.